DR64 Primary FT PCBA Level Requirements

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1 Introduction

1.1 General Introduction

SM-44041 - General Introduction

1. This document outlines the functional test requirements for the Mobileye PCBA level.
2. This document includes:
    a. Testing rules.
    b. Fixture instruction.
    c. interfaces connections.
    d. Host PC orchestrations.
    e. API & GUI requirements.
3. This document is based on core design BOM, test development shall be based on BOM assembly per ME BRD part number.

1.2 Intended Audience

SM-44013 - Intended Audience

- The document is intended for use test development team as UUT functional testing requirements form.

1.3 Acronyms and Terminology

SM-44020 - Acronyms and Terminology Table

Term	Description
FT	**F**unctional **T**est
MCU	**M**icro **C**ontroller **U**nit
UART	**U**niversal **A**synchronous **R**eceiver **T**ransmitter
CAN	**C**ontroller **A**rea **N**etwork
ETH	**E**thernet **I**nterface
PCIe	**P**eripheral **C**omponent **I**nterconnect **E**xpress
BOM	**B**ill **O**f **M**aterials
PN	**P**art **N**umber
BSCAN	**B**oundary **S**can

Term	Description
I/O	**I**nput **O**utput
RefDes	**Ref**erence **Des**ignator
PCBA	**P**rinted **C**ircuit **B**oard **A**ssembly
ME	**M**obil**e**ye
UUT	**U**nit **U**nder **T**est
TAP	**T**est **A**ccess **P**ort

1.4 Related Documents

SM-44022 - Related Documents Table

The table below lists resources of information that are relevant to the current document. Customer-supplied documents are available in the Mobileye SW Bundle Documentation Package

Name	Location
Vobes	Vobes
SV62 HSI	SV62 HSI C
DR64 Primary HSI	DR64 Primary HSI B2

1.5 Roles

SM-44014 - Roles

Responsibility	Contact Person
AV DFT - Team Lead	Nahum Nahumov
Senior Platform Engineering Manager	Alon Rotman

1.6 Revision History

Revision History

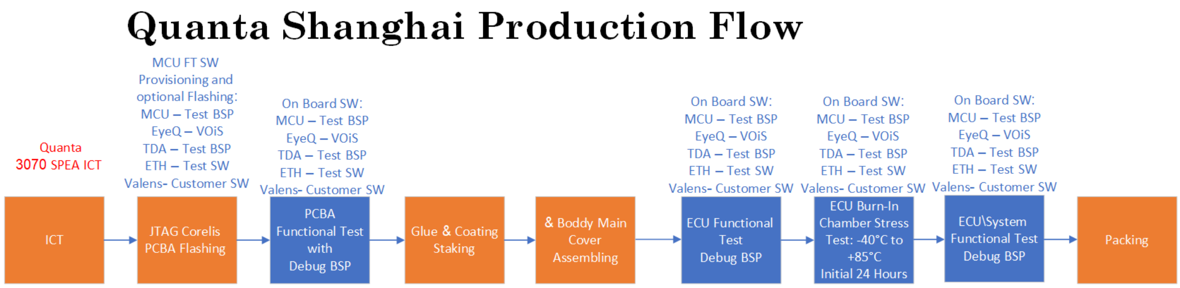
Ver.	Date	Author	Essence of change	Reviewers	Approved by
0.1	29-Sep-2025	Nahum N.	Initial Release	Alon R.	Alon R.

Ver.	Date	Author	Essence of change	Reviewers	Approved by
0.2	9-Dec-2025	Nahum N.	remove wrong requirements" EyeQ U59 & EyeQ ADC 3.7.6.2, 3.7.6.4"	Alon R.	Alon R.
0.3	2-Mar-2026	Nahum N.	Updated production flow process diagrams; removed the Run-in test step	Yaron D.	Yaron D.

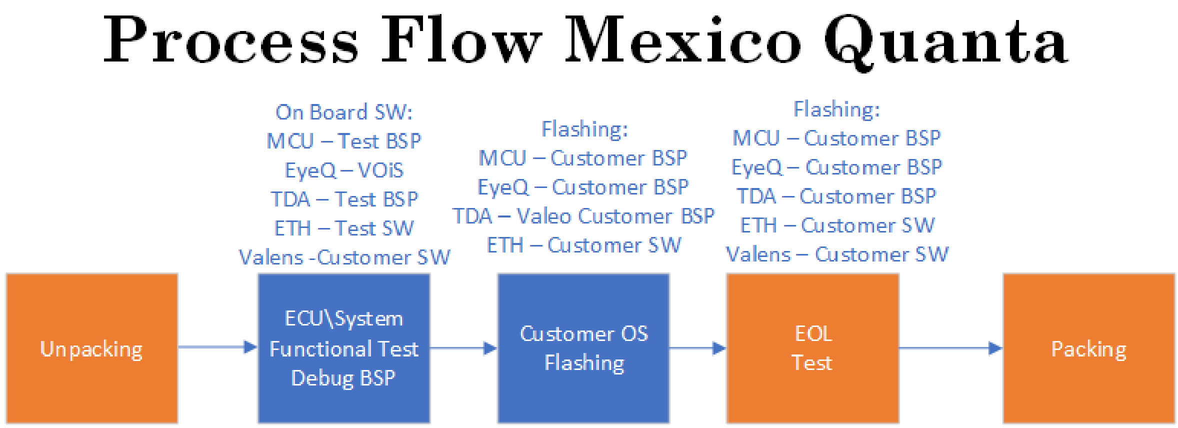
1.7 Production Test Process

Production Test Process

SM-44017 - Quanta Shanghai Production Test Process



SM-44031 - Quanta Mexico Production Test Process



SM-43766 - Each PCBA shall have the test software pre-flashed before executing this test step.

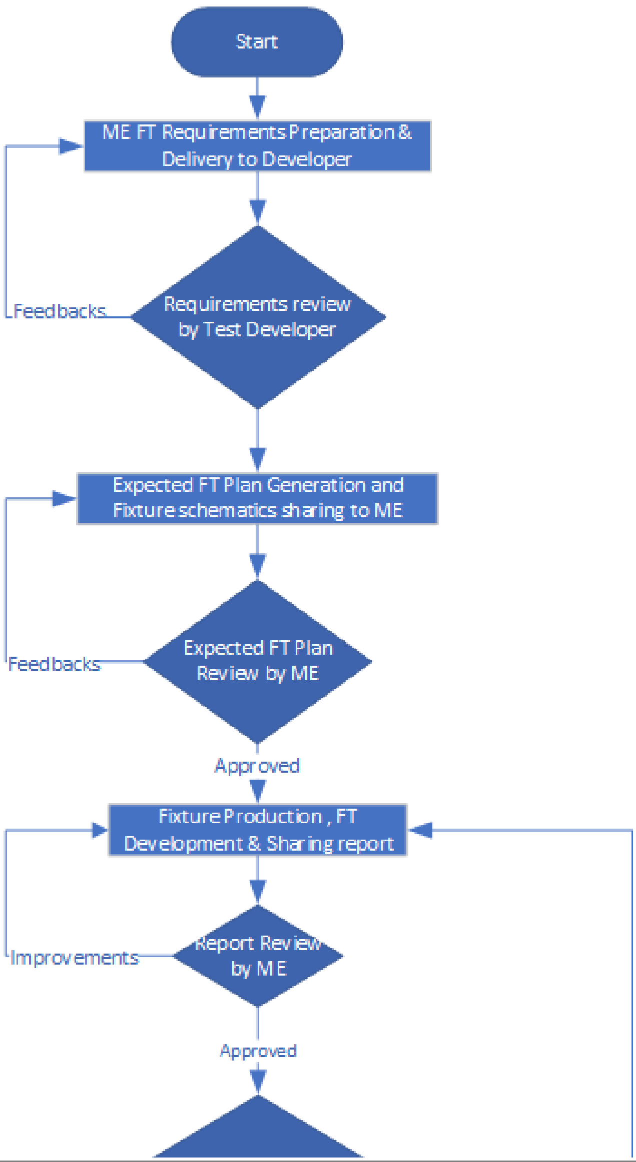
2 General Rules

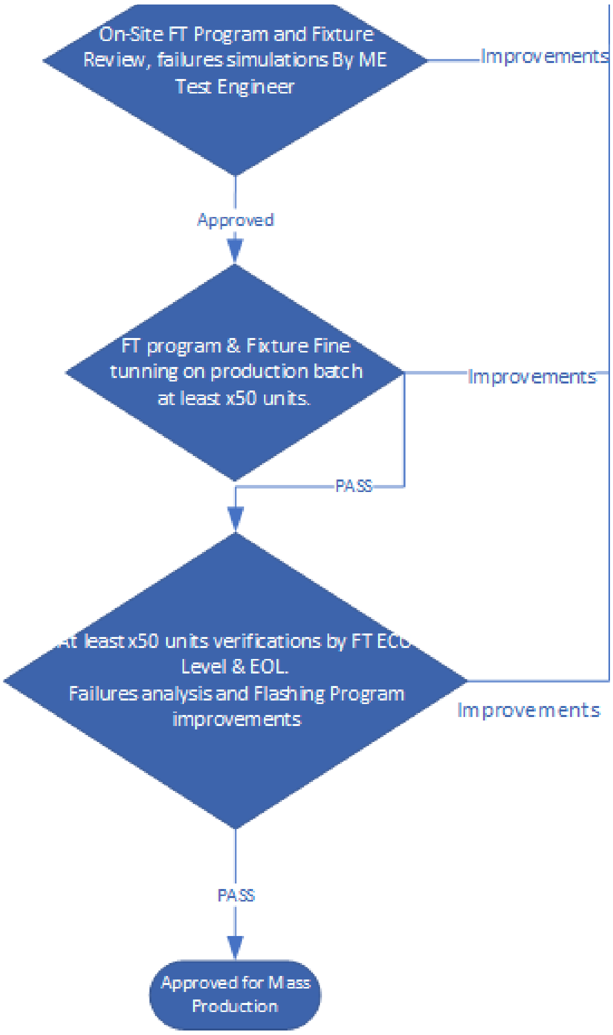
2.1 Test Program and Fixture Approval Process

SM-43735 - Test Program and Fixture Approval Process

1. **Test Program and Fixture Approval Process** Test program development, fixture development, and production activities shall be approved by ME Test Engineering.

2. The FT test program and fixture must be approved in accordance with the approval process outlined in this document before being released to mass production.
3. The below approval process shall be implemented





4. A testing plan and report shall be generated for each external interface and component pin.
5. Testing coverage documentation shall clearly indicate the specific testing types performed on each device to ensure traceability and comprehensive validation.

2.2 PCBA FT General Requirements

SM-43738 - PCBA FT General Requirements

1. **PCBA FT General Requirements** Testing can be performed via UART and SSH connections to the NXP MCU test software through the PCBA's external Ethernet port. This setup enables execution of CLI commands to manage each SoC (e.g., MCU, TDA, EyeQ) over internal UART links between the NXP and the TDA/EyeQ. Required software files can be transferred using TFTP.

2. Testing shall be executed in parallel, to the extent feasible, in order to maximize efficiency.
3. File optimization must be carried out to reduce the duration of testing processes.

4. The fastest available Ethernet port shall be used for file transfers to maximize data transfer speed.
5. Wherever feasible, the Device ID of each component shall be verified and recorded in the test log.
6. Wherever feasible, each component shall be detected and tested.
7. Each SoC (e.g., NXP MCU, TDA4, EyeQ) shall be verified by logging into its operating system (e.g., Linux, MEST) and confirming, at a minimum, the OS version to ensure compatibility with the test sequence.
8. The testing program shall be designed to avoid back-driving and signal direction conflicts.
9. The power on and graceful power off shall be implemented and controlled by test program.

2.3 Testing JIG Requirement

SM-44025 - Testing JIG Requirement

1. Use a mechanical locking mechanism to connect and disconnect the UUT from the fixture.

2. Connect the fixture harness and connectors to the UUT's external interfaces using a mechanical side-access connection mechanism.
3. Additional electronic components should be integrated into the fixture, such as a T1 adapter, CAN bus adapter for ECU wake-up, and other necessary modules.
4. Direct connection to the UUT using additional equipment must comply with the UUT's logic and fan-out characteristics.
5. Fixture shall include labels with: ECU PN & Revision and jig number.
6. The fixture shall include a UUT connection counter to support periodic maintenance based on usage.
7. Connect the power supply sense lines (V+ and V–) as close as possible to the UUT on the fixture connector to ensure accurate voltage compensation.
8. UUT wake-up shall be triggered using a CAN bus adapter (e.g., Kvaser or an equivalent device) connected to the UUT’s CAN wake-up port.
9. External fans should be used to provide cooling for each SoC.

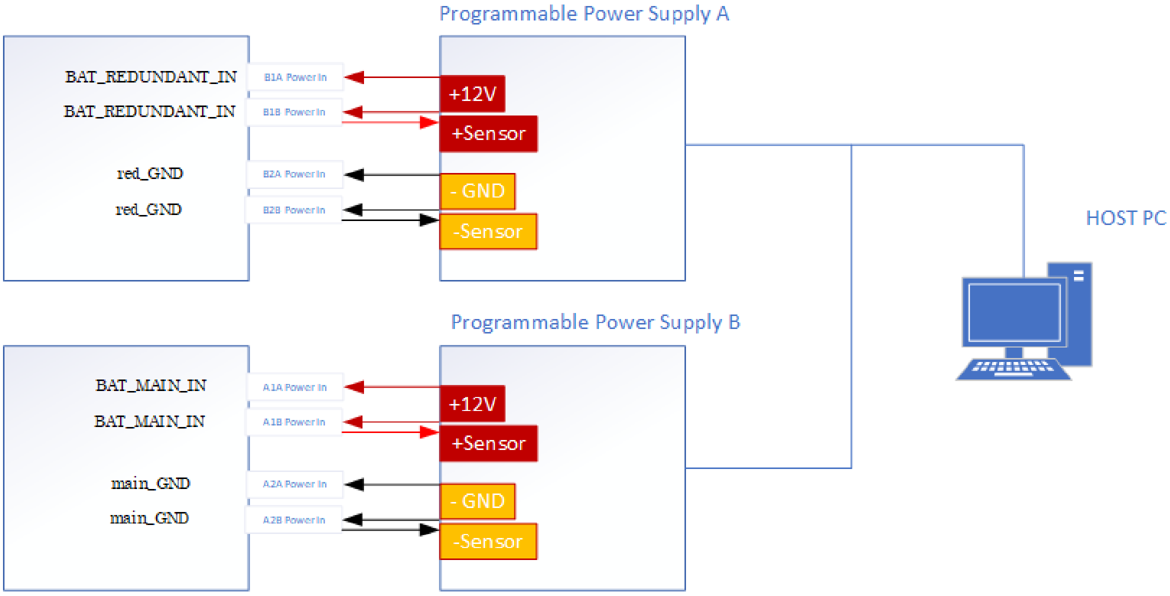
2.4 Cooling Requirement

SM-43756 - Cooling for the PCBA component side shall be provided by three (3) fans, each delivering a minimum airflow of 40 CFM, for a combined total airflow of at least 120 CFM. The cooling airflow shall be directed to each SoC to ensure adequate thermal management.

2.5 Power Test Rules

2.5.1 Power Test Rules

SM-43747 - Power Connection Schematics:



SM-43750 - An external remotely controlled power supply, such as the TDK Z20-40-I, Kikusui, or an equivalent model, shall be used.

SM-44039 - The default state of the power supply shall be OFF.

SM-44042 - The power supply shall be preprogrammed to 12 VDC with a current limit of 5 A before power on.

SM-44036 - The external power supply must have an accuracy of ±1%.

SM-44038 - The host software shall automatically control power ON/OFF as defined by the flashing sequence.

SM-44023 - Verify the UUT power inlet by monitoring the output voltage and current of the external power supply during testing.

SM-44015 - A 12V supply shall be provided to both the main and redundant power inlets of the UUT as follows:

1. Main Power Input: Connect the positive potential (BAT_MAIN_IN) to J21-1 pin B1, and ground (main_GND) to J21-1 pin B2.
2. Redundant Power Input: Connect the positive potential (BAT_REDUNDANT_IN) to J21-2 pin A1, and ground (red_GND) to J21-2 pin A2.

SM-44018 - UUT wake-up shall be triggered using a CAN bus adapter (e.g., Kvaser or an equivalent device) connected to the UUT's CAN wake-up port.

SM-44032 - Monitor voltage and current using connected power sensors or ADCs.

SM-44072 - Graceful Power-Off Control for PCBA

The PCBA shall perform a controlled (graceful) power-off sequence before disconnecting the BAT_MAIN_IN & BAT_REDUNDANT_IN inlets, based on ignition status and CAN wake-up activity, ensuring system integrity and preventing data loss.

Conditions & Actions:

1. **Ignition Cluster Populated**:

    - Drive the ignition signal low to initiate shutdown.
    - If accessible, monitor the MCU (NXP) UART interface for a “power-down” message.
    - Verify that the power supply current is less than 100 µA before disconnecting BAT_MAIN_IN (BAT_REDUNDANT_IN in case populated).

2. **Ignition Cluster Not Populated**:

- Power-off shall only occur when both of the following conditions are met:

    a. No toggling or packet activity is detected for at least 30 seconds on each populated CAN wake-up port.
    b. Power supply current is measured at less than 100 µA.

- Once these conditions are satisfied, disconnect BAT_MAIN_IN and BAT_REDUNDANT_IN in case populated.

2.5.2 The power consumption test described below shall be performed:

SM-44034 - Quiescent current measurement

PCBA	Minimum	Typical	Maximum
Primary	38µA	48µA	100µA

SM-44026 - Operating current measurement in the "startup behavior" operating mode

PCBA	Minimum	Typical	Maximum
Primary	3.6A	4.87A	6.5A

SM-44028 - Operating current measurement in the "shutdown behavior" operating mode

PCBA	Minimum	Typical	Maximum
Primary	0.3A	1.52A	1.85A

SM-44068 - Operating current measurement in the "wakeup behavior" operating mode

PCBA	Minimum	Typical	Maximum
Primary	0.4A	0.5A	0.65A

SM-44070 - Operating current measurement in the operating mode

PCBA	Minimum	Typical	Maximum
Primary	2.25A (36W/16V)	3A	23A (210W/9V)

2.5.3 The Operational, Under-Power, and Over-Power tests described below shall be performed:

SM-44067 - Undervoltage Test

1. **Undervoltage Test** Set ECU inputs voltage to 6V.
2. Check system "Sleep" mode activity by ETH or CAN ports.

SM-44029 - Overvoltage Test

1. Set ECU inputs voltage to 28V.
2. Check system "Sleep" mode activity by ETH or CAN ports.

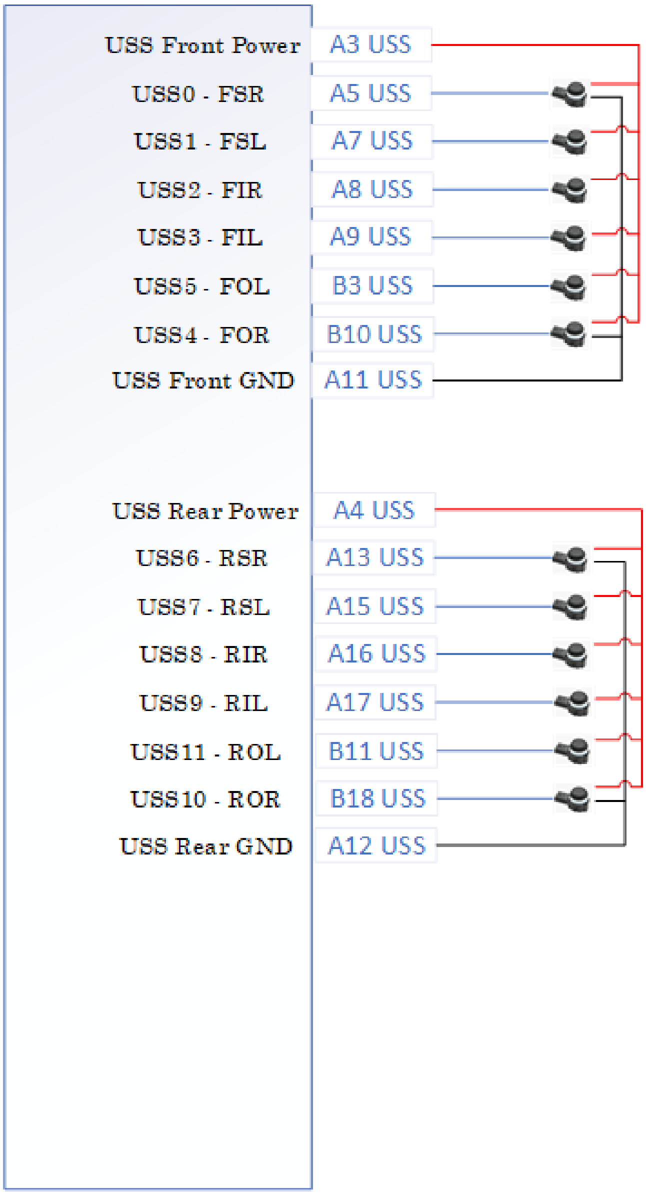
SM-44055 - Operational Test

1. Set ECU inputs voltage to 12V.
2. Check system bootup.

3 SW & HW Test Requirements

3.1 USS Interfaces Test Requirements

SM-44047 - USS connection schematics



SM-44050 - Each USS interface shall be connected to a USS sensor and verified using the test program.

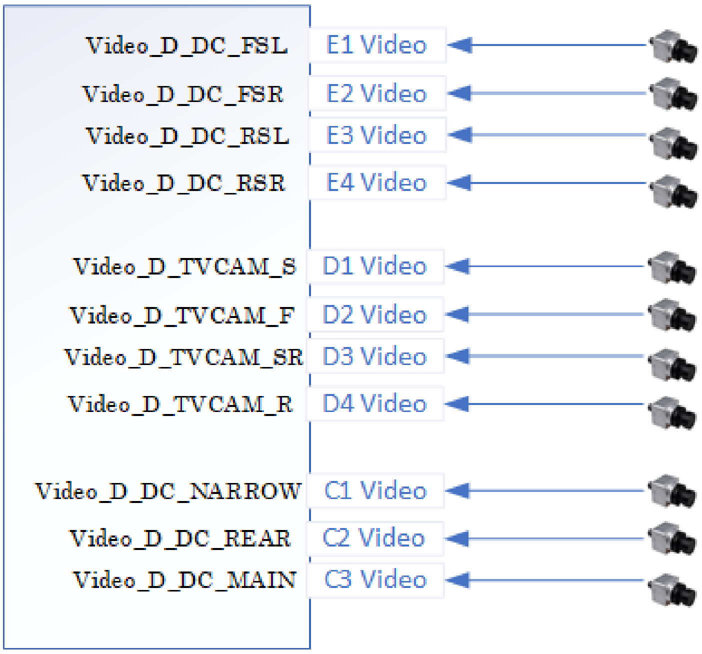
SM-44062 - Each USS power port shall undergo testing for both nominal loading and over-current protection behavior.

SM-44065 - During the test, the USS current shall be measured and verified via the TDA4 ADC.

SM-44058 - Testing shall be performed on the USS UPID ASIC SPI interface and interrupt.

3.2 Video GMSL Interfaces Test Requirement

SM-44060 - Video GMSL Interfaces connection schematics:



SM-43980 - Each Video GMSL Interface shall be connected to a camera and verified using the test program.

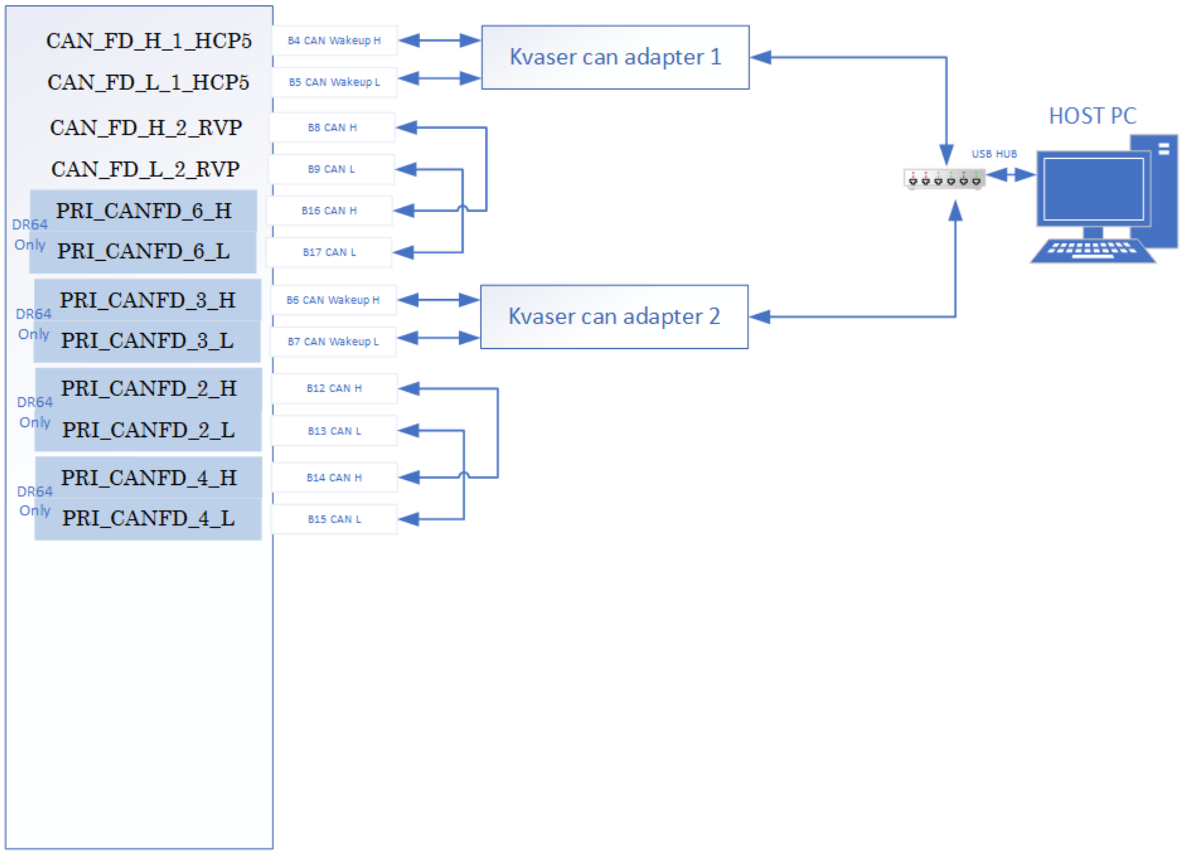
SM-43982 - Each video camera connected to the GMSL interface shall be detected using the I^2C tool, and the camera part number shall be verified.

SM-43963 - For each camera connected via the GMSL interface, image grabbing shall be verified to confirm proper functionality.

SM-43969 - Each Video GMSL Interface port shall undergo testing for both nominal loading and over-current protection behavior.

3.3 CAN Interfaces Test Requirements

SM-43971 - CAN Interfaces connection schematics:



SM-43965 - Each wake-up CAN interface shall be connected to a host PC through a KVASER tool. The interface shall then be verified for correct wake-up behavior by monitoring bus activity and confirming that the expected wake-up signals are received.

SM-43967 - All non-wake-up CAN interfaces shall be verified through loopback testing between ports.

SM-44074 - Each CAN interface with wake-up capability shall be tested by:

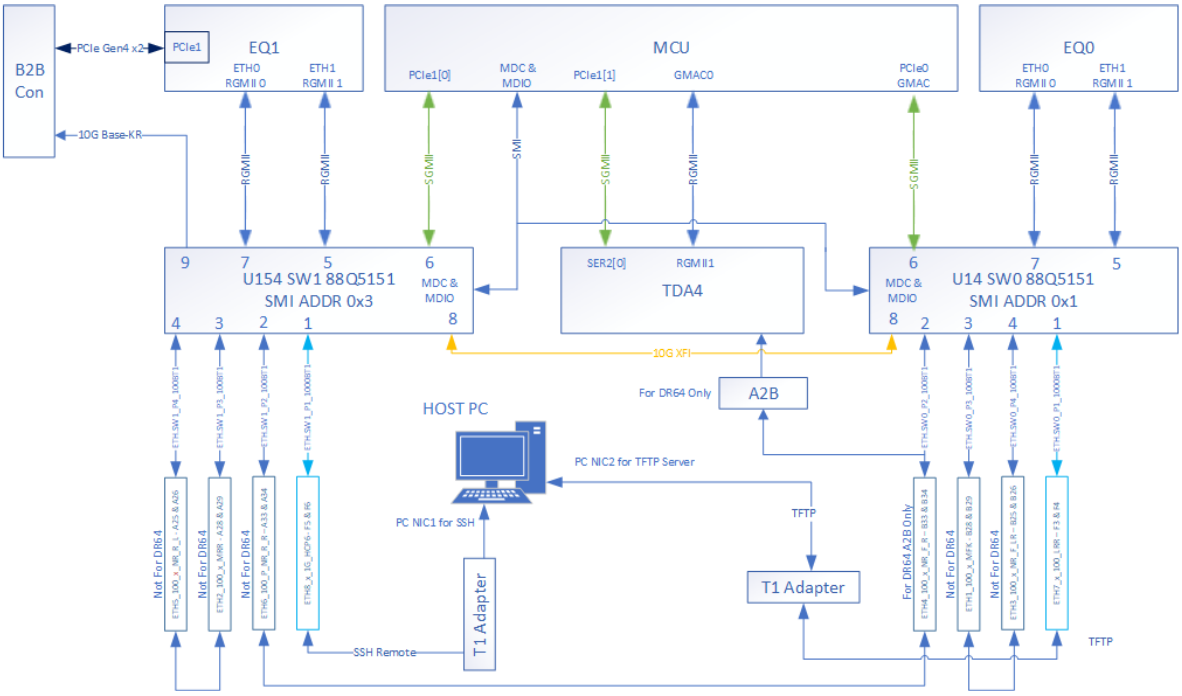
1. Verifying wake-up behavior when a valid CAN frame is received.
2. Transmitting and receiving CAN messages to ensure correct communication.
3. Confirming that all messages are received without errors, data corruption, or loss.
4. No CAN errors (bit, frame, CRC, or acknowledgment errors) shall occur during the test.

SM-43978 - Each CAN interface without wake-up capability shall be tested by:

1. Transmitting and receiving CAN messages to ensure correct communication.
2. Confirming that all messages are received without errors, data corruption, or loss
3. No CAN errors (bit, frame, CRC, or acknowledgment errors) shall occur during the test.

3.4 Ethernet Interfaces Test Requirements

SM-44044 - Ethernet Interfaces connection schematics:



SM-43974 - Ethernet interfaces shall be verified using port-to-port loopback tests or direct connections to the host via a T1 adapter.

SM-44012 - Each Ethernet interface shall be tested using a IPEF, packet generator or TX/RX generator tools with at least 10,000,000 packets in full-duplex mode, and verified to ensure no errors, no packet loss, and no drops on any port. Port statistics can be verified using ethtool.

SM-43987 - Each port rate shall be verified.

SM-43990 - Each port link shall be verified.

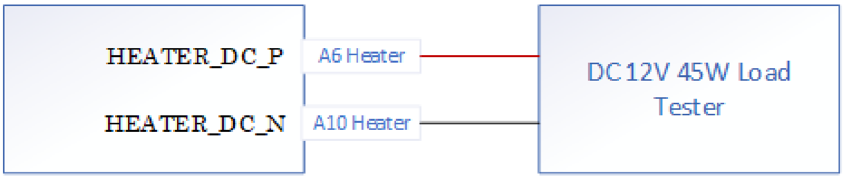
SM-43997 - Ethernet switches shall be managed through the SMI interface to create custom test VLANs.

SM-43999 - each ethernet switch , EyeQ , MCU or TDA ethernet port (RGMII, GMAC, PCIe, XFI, ETH) must be tested

3.5 Heater Interfaces Test Requirements

3.5.1 Heater Interface Connection

SM-44045 - Heater Interfaces connection schematics:



SM-43994 - Heater Interface port shall undergo testing for both nominal loading and over-current protection behavior by external load tester.

SM-44007 - The heater interface shall support VBAT 6-16V DC a maximum load **45W ± 5%**

3.5.2 Nominal load requirement:

SM-44010 - The heater interface shall demonstrate proper nominal-current behavior when subjected to an nominal-current condition simulated by the external load tester.

SM-44002 - Apply a 20 W load to the heater interface.

SM-44005 - The heater interface output voltage shall remain within the range of 11.4 V to 12.6 V when operating under nominal load conditions.

SM-43944 - Measure the voltage at the MCU ADC pin A19, labeled `HEATER.HSS_MON`

3.5.3 Over-current protection requirement.

SM-43942 - The heater interface shall demonstrate proper over-current protection behavior when subjected to an over-current condition simulated by the external load tester.

SM-43943 - Apply a 46W load to the heater interface

SM-43947 - The heater interface output voltage shall not exceed 300 mV when operating under overcurrent load conditions.

SM-43948 - The maximum duration of overcurrent shall not exceed 3 seconds.

SM-43945 - Release the overcurrent load, wait 3 ms, and then repeat the nominal load requirement test.

SM-43946 - After removal of the over-current condition, the heater interface shall automatically resume nominal operation.
Recovery shall occur within **≤ 2 s** after fault removal.

3.5.4 Control Pins

SM-43950 - Each control, status pin must be tested

3.6 MCU NXP 32G3 Requirements

3.6.1 MCU NXP 32G3 BSP SW Requirements

The BSP shall provide the following:

3.6.2 MCU NXP Networking capabilities and tools:

3.6.2.1 I2C Communication

SM-43951 - The BSP shall include an I2C communication tool (I2C tool) to allow reading and writing to I2C devices.

3.6.2.2 Ethernet Interface Management

SM-43949 - The BSP shall include ethtool for querying and controlling Ethernet device parameters.

SM-43954 - The BSP shall include phytool for accessing and modifying Ethernet PHY registers.

3.6.2.3 File Transfer Protocols

SM-43955 - The BSP shall support FTP client and server functionality.

SM-43952 - The BSP shall support TFTP client and server functionality.

3.6.2.4 Network Configuration

SM-43953 - The BSP shall include ifconfig (or equivalent ip command) for network interface configuration.

SM-43958 - The BSP shall support both static IP configuration and DHCP client operation.

3.6.2.5 Network Performance Testing

SM-43959 - The BSP shall include iperf for measuring TCP and UDP throughput.

SM-43956 - The BSP shall include Pktgen for network stress testing.

SM-43957 - The BSP shall TX/RX Gen for network testing.

3.6.2.6 Remote Access

SM-43960 - The BSP shall support SSH server functionality for secure remote shell access.

3.6.2.7 Ethernet Switch Management

SM-43961 - The BSP shall support Ethernet switch SMI/MDIO management.

SM-43853 - The BSP shall include the MCLI tool for switch configuration and monitoring.

3.6.3 PCIe

SM-43857 - The BSP shall include lspci for PCIe device verification, listing, and details.

3.6.4 Device Tree Board Configuration

SM-43893 - The BSP shall include board specific pin mux and peripheral configuration

3.6.5 MCU NXP Serial Communication Requirements

SM-43897 - The BSP shall support UART console access to each ECU SOC such EyeQ, TDA4.

SM-43888 - The BSP shall include minicom for serial communication.

SM-43891 - The BSP shall include at least one alternative UART verified terminal application (e.g., picocom or screen).

3.6.6 MCU NXP File System Tolls Requirements

SM-43904 - The BSP shall include md5sum for file integrity verification.

SM-43907 - The BSP shall include resize2fs for resizing ext2/ext3/ext4 file systems.

SM-43899 - The BSP shall include dd for raw data read/write operations to storage devices.

3.6.7 MCU NXP SoC & Hardware Control Requirements

SM-43902 - The BSP shall provide SoC control utilities for managing clock, power, and reset functions.

SM-43885 - The BSP shall include dmidecode for retrieving hardware/system information.

SM-43879 - The BSP shall include dmesg for accessing kernel ring buffer logs.

SM-43881 - The BSP shall include pktgen for generating network packets.

SM-43794 - The BSP shall include an SPI communication tool for accessing SPI devices.

SM-43796 - The system shall support graceful power-off and provide a confirmation message upon successful shutdown.

3.6.8 MCU NXP Bootloader & Kernel Requirements

SM-43790 - The BSP shall provide access to the bootloader (e.g., U-Boot) for environment configuration and recovery.

SM-43792 - The BSP shall support kernel message retrieval.

SM-43799 - The BSP shall correctly handle and display ASCII and Unicode text characters in console output.

SM-43801 - Text character strings in the BSP shall be free of colors, tables, and graphics to prevent corrupted strings in the automation GUI.

SM-43798 - Monitoring interruptions are not allowed; each measurement shall be executed only upon request.

3.6.9 MCU NXP 32G3 Cluster Test Requirement

The MCU NXP BSP shall provide support for, and include, the tests listed below:

3.6.9.1 MCU NXP Boot Mode strapping

SM-43784 - The NXP MCU boot mode strapping signals BOOTMOD0 and BOOTMOD1 shall be controllable, when required, by the test program and test fixture through direct access to the corresponding test points on the PCBA.

SM-43787 - If the resistor BOM configuration sets the default boot mode to boot from the programmed NOR flash and eMMC (with OTP e-Fusing) by pulling BOOTMOD0 = 0 and BOOTMOD1 = 0, then, prior to OTP e-Fusing, BOOTMOD0 shall be driven to logic high before MCU boot-up or reset.

SM-43776 - MCU NXP datasheet boot modes table

Life cycle state	FUSE_SEL	BOOTMOD1	BOOTMOD2	Mode	Use case
CUST_DEL or OEM_PROD or IN_FIELD	0	0	0	Serial boot + XOSC in Differential mode	Production configuration for an unprogrammed chip
	0	0	1	Serial boot + XOSC in Crystal mode or Bypass mode	Production configuration for an unprogrammed chip
	0	1	0	Boot from RCON	Development configuration
	1	0	0	Boot from fuses	Production configuration for a programmed chip
	1	0	1	Boot from fuses	Production configuration for a programmed chip
	1	1	0	Serial boot	Debug of a programmed chip
	X	1	1	Reserved	Reserved

3.6.9.2 NOR Flash Test BSP Requirements

SM-43780 - NOR Flash Detection

The BSP shall detect SPI NOR flash as /dev/mtdblock and display detailed information about the NOR Flash.

SM-43834 - NOR Flash Manufacturer & Device ID Detection

The BSP shall detect and display the manufacturer and device ID codes of the NOR Flash.

SM-43837 - NOR Flash Size

The BSP shall detect and display the size of the NOR Flash.

SM-43828 - NOR Flash Register Read/Write

The BSP shall support reading from and writing to NOR Flash registers.

SM-43831 - NOR Flash Dump/Upload

The BSP shall support dumping NOR Flash contents to a binary file and uploading from a binary file to NOR Flash.

3.6.9.3 eMMC Requirements

SM-43841 - eMMC Detection

The BSP shall detect eMMC as /dev/mmcblock and display detailed information about the eMMC.

SM-43844 - eMMC Size

The BSP shall detect and display the size of the eMMC.

SM-43808 - eMMC Register Read/Write

The BSP shall support reading from and writing to eMMC registers.

SM-43821 - eMMC Dump/Upload

The BSP shall support dumping eMMC contents to a binary file and uploading from a binary file to eMMC.

3.6.9.4 U59 EEPROM Requirements

SM-43825 - U59 EEPROM Register Read/Write

The BSP shall support reading from and writing to U59 EEPROM registers.

SM-43813 - U59 EEPROM Dump/Upload

The BSP shall support dumping U59 EEPROM contents to a binary file and uploading from a binary file to U59 EEPROM.

SM-43817 - U59 EEPROM Write Protection

The BSP shall support enabling and disabling write protection for U59 EEPROM.

3.6.9.5 MCU GPIOs & Interrupts Control

SM-43725 - MCU GPIOs & Interrupts Control

The BSP shall support setting and reading values of MCU GPIOs and controlling interrupts

3.6.9.6 Ethernet Switch Management

SM-43728 - ETH SW MDIO Management

The BSP shall support Ethernet switch SMI/MDIO management.

3.6.9.7 MCU ADC Requirements

SM-43719 - MCU ADC Measurement

The BSP shall support measurement of each A2D channel in the MCU ADC

3.6.9.8 CAN Communication Requirements

SM-43722 - CAN Communication

The BSP shall support CAN communication to secondary control and traffic monitoring.

SM-43731 - CANFD Communication

The BSP shall support CANFD communication to car control, wakeup, and traffic monitoring.

3.6.9.9 LPDDR Memory Requirements

SM-43696 - LPDDR Memory Test

The system shall support LPDDR memory testing using memtester.

SM-43699 - To reduce the functional test duration, it is permitted to stop the LPDDR memtest after successfully completing test step 6

3.7 EyeQ Requirements

3.7.1 EyeQ Test OS Requirements

SM-43712 - Both EyeQ functional test shall utilize either VOiS, DV MEST, or an equivalent approved alternative.

The selected EyeQ Test OS shall provide the following :

3.7.2 EyeQ Networking capabilities and tools:

3.7.2.1 I2C Communication

SM-43716 - The BSP shall include an I2C communication tool (I2C tool) to allow reading and writing to I2C devices.

3.7.2.2 Ethernet Interface Management

SM-43706 - The BSP shall include ethtool for querying and controlling Ethernet device parameters.

SM-43709 - The BSP shall include phytool for accessing and modifying Ethernet PHY registers.

3.7.2.3 File Transfer Protocols

SM-43762 - The BSP shall support FTP client and server functionality.

SM-43764 - The BSP shall support TFTP client and server functionality.

3.7.2.4 Network Configuration

SM-43759 - The BSP shall include ifconfig (or equivalent ip command) for network interface configuration.

SM-43761 - The BSP shall support both static IP configuration and DHCP client operation.

3.7.2.5 Network Performance Testing

SM-43745 - The BSP shall include iperf for measuring TCP and UDP throughput.

SM-43737 - The BSP shall include Pktgen for network stress testing.

SM-43740 - The BSP shall TX/RX Gen for network testing.

3.7.2.6 Remote Access

SM-43754 - The BSP shall support SSH server functionality for secure remote shell access.

3.7.2.7 PCIe

SM-43758 - The BSP shall include lspci for PCIe device verification, listing, and details.

3.7.2.8 Device Tree Board Configuration

SM-43749 - The BSP shall include board specific pin mux and peripheral configuration

3.7.3 EyeQ File System Tolls Requirements

SM-43752 - The BSP shall include md5sum for file integrity verification.

SM-44040 - The BSP shall include resize2fs for resizing ext2/ext3/ext4 file systems.

SM-44043 - The BSP shall include dd for raw data read/write operations to storage devices.

3.7.4 EyeQ Hardware Control Requirements

SM-44037 - The BSP shall provide SoC control utilities for managing clock, power, and reset functions.

SM-44021 - The BSP shall include dmidecode for retrieving hardware/system information.

SM-44024 - The BSP shall include dmesg for accessing kernel ring buffer logs.

SM-44016 - The BSP shall include pktgen for generating network packets.

3.7.5 EyeQ Bootloader & Kernel Requirements

SM-44019 - The BSP shall provide access to the bootloader (e.g., U-Boot) for environment configuration and recovery.

SM-44033 - The BSP shall support kernel message retrieval.

SM-44035 - The BSP shall correctly handle and display ASCII and Unicode text characters in console output.

SM-44027 - Text character strings in the BSP shall be free of colors, tables, and graphics to prevent corrupted strings in the automation GUI.

SM-44030 - Monitoring interruptions are not allowed; each measurement shall be executed only upon request.

3.7.6 EyeQ BSP Test Support

The EyeQ BSP shall provide support for, and include, the tests listed below:

3.7.6.1 eMMC Tests

SM-44069 - The EyeQ BSP shall detect the eMMC as /dev/mmcblock and display detailed information about the eMMC.

SM-44071 - The EyeQ BSP shall provide functionality to read the eMMC size.

SM-44046 - The EyeQ BSP shall support register read and write operations for the eMMC.

SM-44053 - The EyeQ BSP shall support dumping and uploading eMMC data to and from a binary file.

3.7.6.2 EyeQ GPIOs & Interrupts

SM-44063 - The EyeQ BSP shall support setting and reading values of MCU GPIOs and interrupts.

3.7.6.3 CAN & CANFD

SM-44059 - The EyeQ BSP shall support CAN communication to secondary control and traffic handling.

SM-44061 - The EyeQ BSP shall support CANFD communication to car control, wakeup, and traffic handling.

3.7.6.4 LPDDR Memory

SM-43981 - The EyeQ BSP shall support LPDDR memory testing using memtester.

SM-43962 - To reduce the functional test duration, it is permitted to stop the LPDDR memtest after successfully completing test step 6

3.7.6.5 MIPI Interfaces

SM-43964 - The EyeQ BSP shall support image grabbing from MIPI In 0 and 1 camera interfaces to VDI1 and VDI2.

SM-43970 - The EyeQ BSP shall support MIPI DPHY output to TDA testing.

3.7.6.6 SPI to VDI

SM-43972 - The EyeQ BSP shall support reading and writing to VDI1 and VDI2 via SPI tools.

3.7.6.7 Lidar

SM-43966 - The EyeQ BSP shall support Lidar testing.

3.8 TDA4 Requirements

3.8.1 TDA4 BSP SW Requirements

The BSP shall provide the following:

3.8.1.1 TDA4 Networking capabilities and tools:

SM-43968 - I2C Communication

The BSP shall include an I2C communication tool (I2C tool) to allow reading and writing to I2C devices.

3.8.1.2 Ethernet Interface Management

SM-43977 - The BSP shall include ethtool for querying and controlling Ethernet device parameters.

SM-43979 - The BSP shall include phytool for accessing and modifying Ethernet PHY registers.

3.8.1.3 File Transfer Protocols

SM-43973 - The BSP shall support FTP client and server functionality.

SM-43975 - The BSP shall support TFTP client and server functionality.

3.8.1.4 Network Configuration

SM-43992 - The BSP shall include ifconfig (or equivalent ip command) for network interface configuration.

SM-43988 - The BSP shall support both static IP configuration and DHCP client operation.

3.8.1.5 Network Performance Testing

SM-43991 - The BSP shall include iperf for measuring TCP and UDP throughput.

SM-43998 - The BSP shall include Pktgen for network stress testing.

SM-44000 - The BSP shall TX/RX Gen for network testing.

3.8.1.6 Remote Access

SM-43993 - The BSP shall support SSH server functionality for secure remote shell access.

3.8.1.7 Device Tree Board Configuration

SM-43995 - The BSP shall include board specific pin mux and peripheral configuration

3.8.1.8 TDA4 File System Tolls Requirements

SM-44008 - The BSP shall include md5sum for file integrity verification.

SM-44011 - The BSP shall include resize2fs for resizing ext2/ext3/ext4 file systems.

SM-44003 - The BSP shall include dd for raw data read/write operations to storage devices.

3.8.1.9 TDA4 SoC & Hardware Control Requirements

SM-43916 - The BSP shall provide SoC control utilities for managing clock, power, and reset functions.

SM-43932 - The BSP shall include dmidecode for retrieving hardware/system information.

SM-43933 - The BSP shall include dmesg for accessing kernel ring buffer logs.

SM-43930 - The BSP shall include pktgen for generating network packets.

SM-43931 - The BSP shall include an SPI communication tool for accessing SPI devices.

SM-43938 - The system shall support graceful power-off and provide a confirmation message upon successful shutdown.

3.8.1.10 TDA4 Bootloader & Kernel Requirements

SM-43940 - The BSP shall provide access to the bootloader (e.g., U-Boot) for environment configuration and recovery.

SM-43935 - The BSP shall support kernel message retrieval.

SM-43936 - The BSP shall correctly handle and display ASCII and Unicode text characters in console output.

SM-43941 - Text character strings in the BSP shall be free of colors, tables, and graphics to prevent corrupted strings in the automation GUI.

SM-43926 - Monitoring interruptions are not allowed; each measurement shall be executed only upon request.

3.8.2 TDA4 Cluster Test Requirement

3.8.2.1 TDA4 MCU NXP BSP Test Support

The TDA4 NXP BSP shall provide support for, and include, the tests listed below:

3.8.2.2 TDA4 NOR Flash Tests

SM-43928 - The TDA4 NXP BSP shall detect the SPI NOR flash as /dev/mtdblock and display detailed information about the NOR flash.

SM-43865 - The TDA4 NXP BSP shall detect and display the manufacturer and device ID code of the NOR flash.

SM-43866 - The TDA4 NXP BSP shall provide functionality to read the NOR flash size.

SM-43863 - The TDA4 NXP BSP shall support register read and write operations for the NOR flash.

SM-43864 - The TDA4 NXP BSP shall support dumping and uploading NOR flash data to and from a binary file.

3.8.2.3 TDA4 eMMC Tests

SM-43872 - The TDA4 NXP BSP shall detect the eMMC as /dev/mmcblock and display detailed information about the eMMC.

SM-43874 - The TDA4 NXP BSP shall provide functionality to read the eMMC size.

SM-43868 - The TDA4 NXP BSP shall support register read and write operations for the eMMC.

SM-43870 - The TDA4 NXP BSP shall support dumping and uploading eMMC data to and from a binary file.

3.8.2.4 TDA4 U128 EEPROM Tests

SM-43862 - The TDA4 NXP BSP shall support register read and write operations for the U128 EEPROM.

SM-43855 - The TDA4 NXP BSP shall support dumping and uploading U128 EEPROM data to and from a binary file.

SM-43859 - The TDA4 NXP BSP shall support enabling and disabling write protection for the U128 EEPROM.

3.8.2.5 TDA4 GPIOs & Interrupts

SM-43895 - The TDA4 NXP BSP shall support setting and reading values of GPIOs and interrupts.

3.8.2.6 TDA4 MCU ADC

SM-43898 - The TDA4 NXP BSP shall support measurement of each MCU A2D channel.

3.8.2.7 TDA4 LPDDR Memory

SM-43889 - The TDA4 NXP BSP shall support LPDDR memory testing using memtester.

3.8.2.8 TDA4 CSI / DPHY Interfaces

SM-43892 - The TDA4 NXP BSP shall support DPHY (TFL) CSI2 TX testing for logging purposes.

SM-43906 - The TDA4 NXP BSP shall support TDA4 + EyeQ Main CSI0 test solution.

SM-43909 - The TDA4 NXP BSP shall support DPHY RAW12 CSI1 RX testing from x4 parking 96724F Deserializer.

3.8.2.9 TDA4 USS

SM-43901 - The TDA4 NXP BSP shall support USS test solution.

3.9 Current Sensors

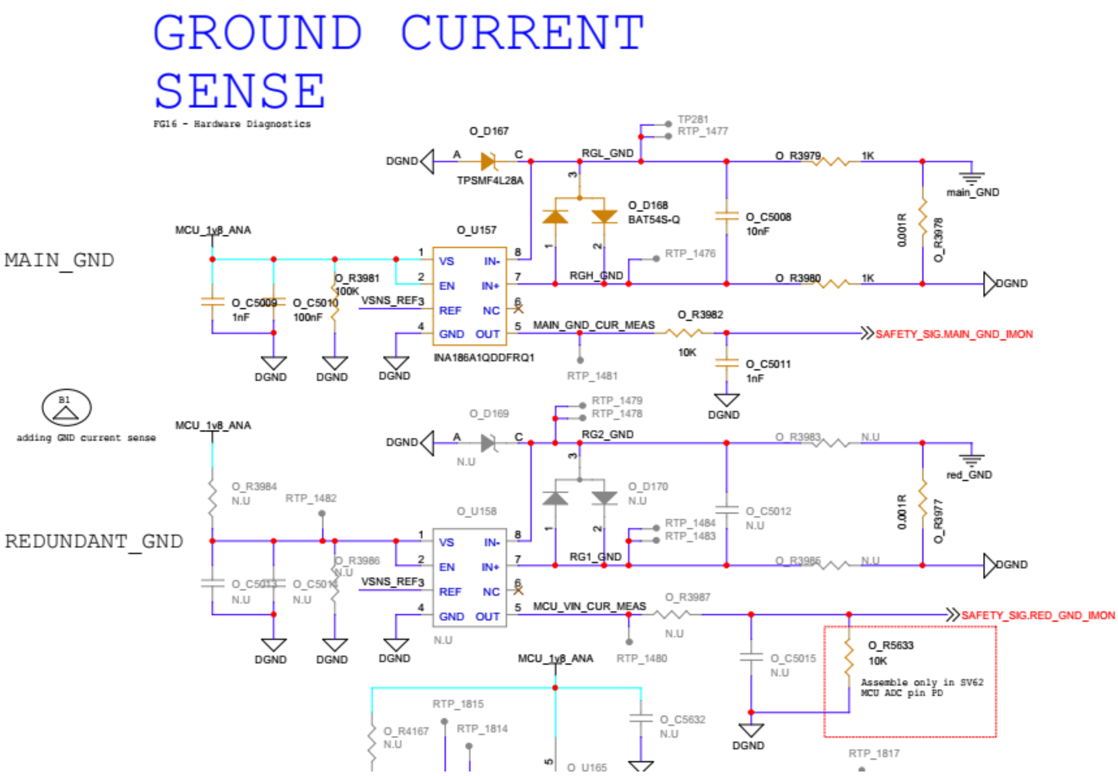
3.9.1 Voltage & Ground Current Sensors

SM-43884 - Voltage & Ground Current Sensors

1. Each current sensor shall be verified by measuring its output voltage using the NXP MCU’s ADC

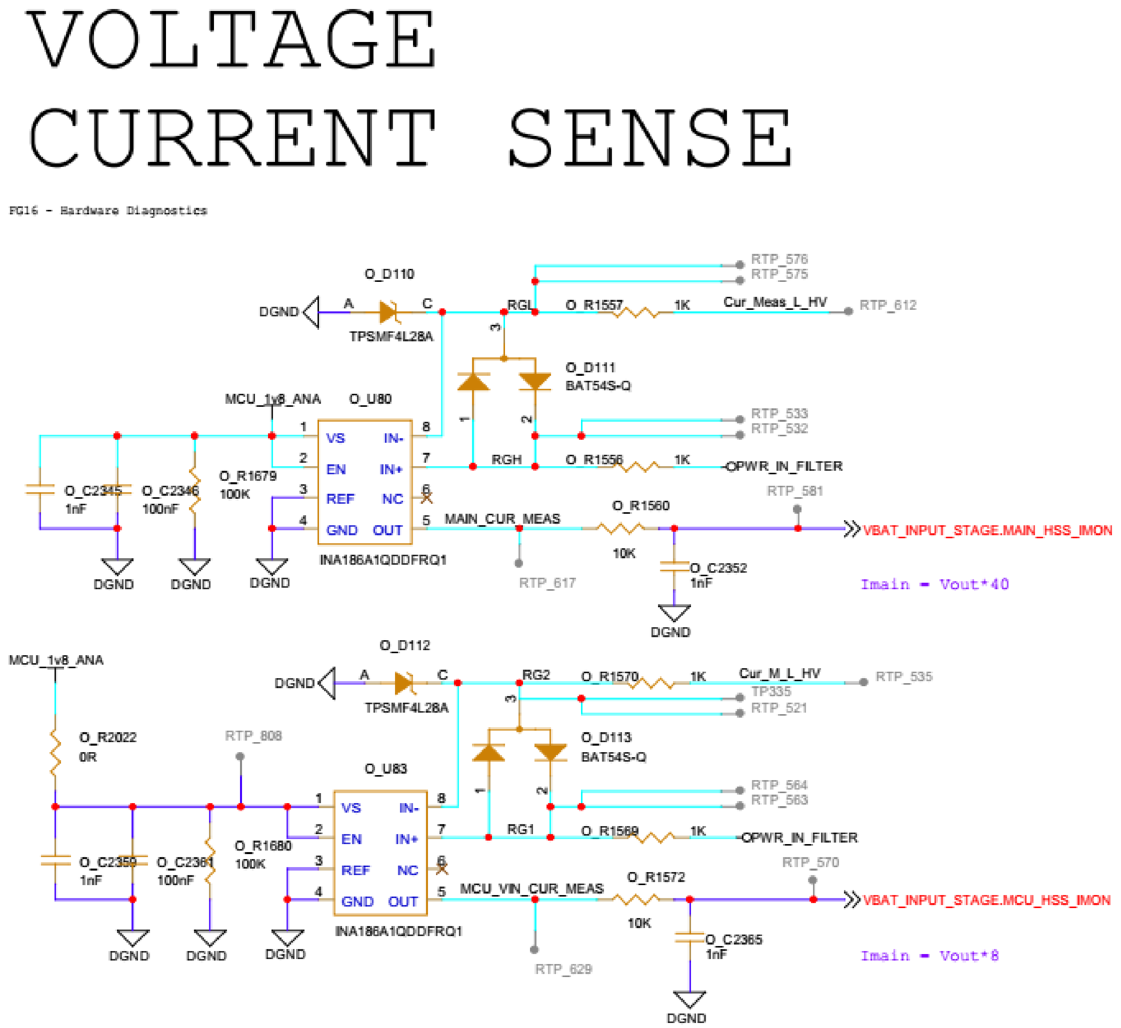
3.9.2 Ground Current Sensors

SM-43886 - Ground Current Sensors schematics



3.9.3 Voltage Current Sensors

SM-43880 - Voltage Current Sensors schematics



3.10 I2C Devices Test Requirements

3.10.1 MCU I2C Devices Test Requirements

SM-43795 - Each I2C device must be tested for acknowledgement.

SM-43797 - Each I2C device must be tested for it's ID code where applicable.

SM-43791 - Each I2C device must be tested for its interrupts and alerts where applicable.

SM-43793 - The quantity of I2C devices must be detected and cross-checked with the BOM

SM-43800 - Each I2C device shall be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry

3.11 I2C Voltage Monitoring Test Requirements

SM-43802 - Each voltage monitor must be tested for acknowledgement.

SM-43772 - The quantity of voltage monitors must be detected and cross-checked with the BOM.

SM-43785 - Each monitored channel reports voltage must be verifiyed within the tolerance specified in the SV62 HSI documentation.

SM-43789 - Each voltage monitor device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry.

SM-43778 - The voltage monitors interrupt pin must be tested to ensure it reliably detects both undervoltage and overvoltage events

3.12 I2C GPIO Extender Test Requirements

SM-43782 - Each GPIO extender must be tested for acknowledgement.

SM-43836 - The quantity of GPIO extenders must be detected and cross-checked with the BOM.

SM-43839 - Each port shall be defined according to the electrical schematics and verified against its part number, revision level, and corresponding BOM entry.

SM-43830 - The GPIO extender’s interrupt pin must be tested to verify that it reliably detects intended events

SM-43833 - Each GPIO extender device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry.

SM-43843 - Each GPIO extender reset pin must be tested.

3.13 I2C Temperature Sensor Test Requirements

SM-43807 - Each I2C temperature sensor must be tested for acknowledgement.

SM-43809 - The quantity of I2C temperature sensors must be detected and cross-checked with the BOM.

SM-43822 - The reading from each I2C temperature sensor must be verified to ensure it is within the tolerance specified in the SV62 HSI.

SM-43826 - Each I2C temperature sensor device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry.

SM-43814 - The I2C temperature sensor alert pin must be tested to ensure it reliably detects out-of-range events.

3.14 I2C Humidity Sensor Test Requirements

SM-43818 - Each I2C humidity sensor must be tested for acknowledgement.

SM-43726 - The quantity of I2C humidity sensors must be detected and cross-checked with the BOM.

SM-43729 - The reading from each I2C humidity sensor must be verified to ensure it is within the tolerance specified in the SV62 HSI.

SM-43720 - Each I2C humidity sensor device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry.

SM-43723 - The I2C humidity sensor alert pin must be tested to ensure it reliably detects out-of-range events.

3.15 I2C EEPROM Test Requirements

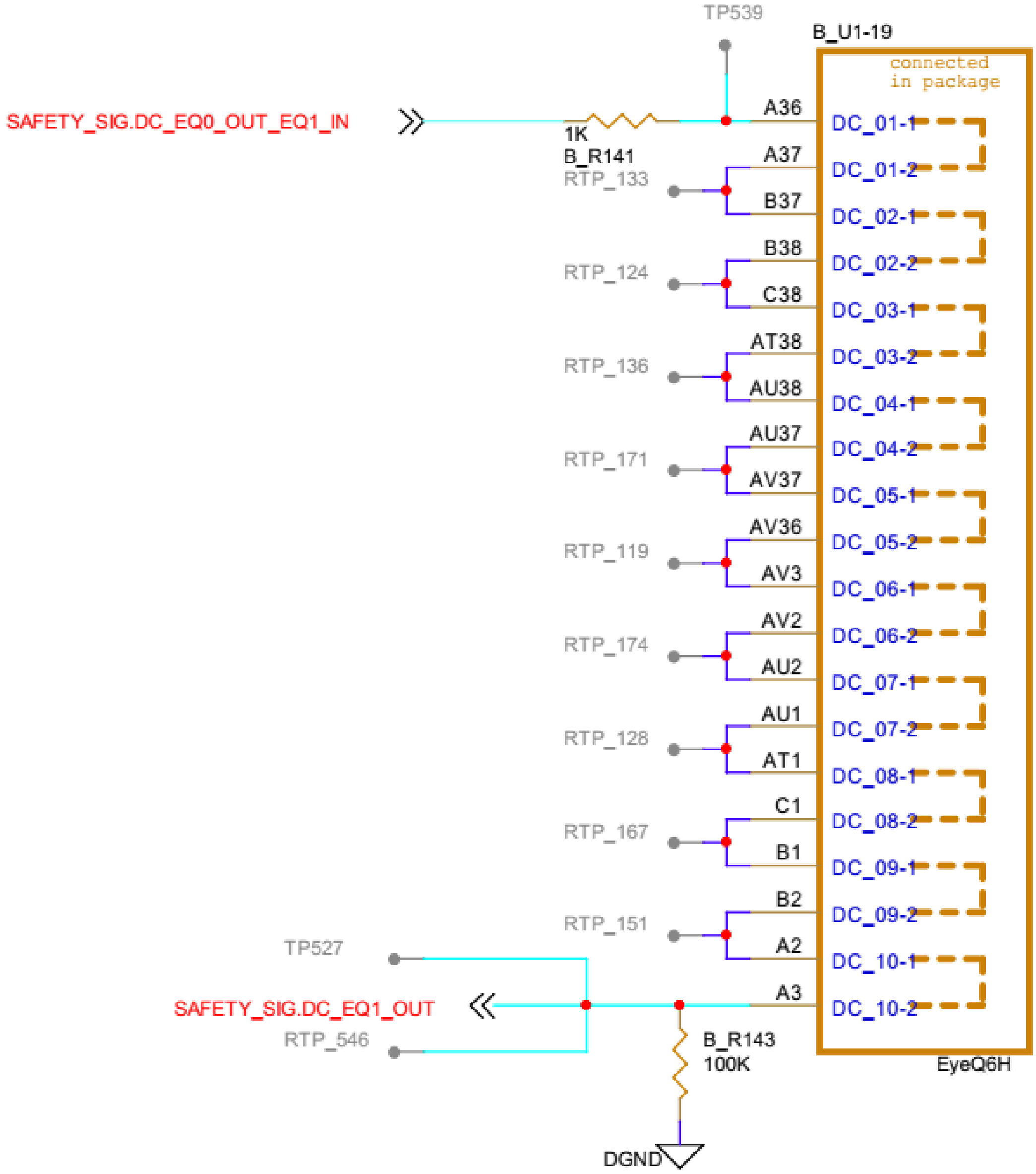
SM-43704 - Each I2C EEPROM device must be tested for acknowledgement.

SM-43697 - The quantity of I2C EEPROM must be detected and cross-checked with the BOM.

SM-43700 - Each I2C EEPROM device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry.

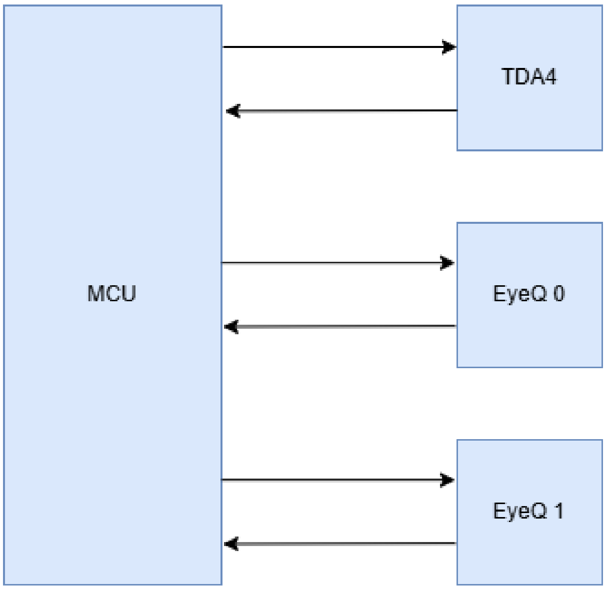
3.16 EyeQ6 Daisy Chain verification Requirement

SM-43713 - The EyeQ6 daisy chain must be verified by driving high and low logic levels on pin A36 and confirming each corresponding logic level on pin A3.



3.17 MCU UART's

SM-43717 - MCU UART's Diagram



SM-43707 - MCU UART's Connection Table

#	Peripheral	Signal	Routed pin/signal	Label	
G10	LLCE_LIN_2	txd	[G10] PC_05	ENC.3v3_UART_RX_R	TDA4
A14	LLCE_LIN_2	rxd	[A14] PC_06	ENC.3v3_UART_TX_R	
F10	LLCE_LIN_0	txd	[F10] PK_15	EQ_IOs.EQ0_3v3_GPIO_A2_UART_0_RX_R	EyeQ_0 (A)
A12	LLCE_LIN_0	rxd	[A12] PL_00	EQ_IOs.EQ0_3v3_GPIO_A3_UART_0_TX_R	EyeQ_0 (A)
B9	LLCE_LIN_3	txd	[B9] PC_07	EQ_IOs.EQ1_3v3_GPIO_A2_UART_0_RX_R	EyeQ_1 (B)
G13	LLCE_LIN_3	rxd	[G13] PL_07	EQ_IOs.EQ1_3v3_GPIO_A3_UART_0_TX_R	EyeQ_1 (B)

SM-43710 - All MCU internal UART interfaces connected to any SoC (e.g., EyeQ, TDA4) shall be verified for both transmit (TX) and receive (RX) functionality by ensuring accurate data transmission and reception without errors.

3.18 MCU Ethernet Switches SMI Port

SM-43763 - SMI Connection Table

Peripheral	Signal	Pin/Port	Label
PFE_MAC0	mdc	[M19] PF_00	MCU_IF.1v8_MDC
PFE_MAC0	md	[L22] PF_01	MCU_IF.1v8_MDIO

SM-43765 - SMI Devices List

Device	P.N	Ref. Des.	Address	Notes
ETH. SW_0	88Q5151	U14	0x01	
ETH. SW_1	88Q5151	U154	0x03	
10G PHY	MVQ3244	U150	0x09	U154 Slave, NA DR64

SM-43760 - All devices connected to the MCU SMI must undergo functionality testing to confirm reliable, error-free data transmission and reception.

SM-43743 - The quantity of SMI supported devices must be detected and cross-checked with the BOM.

SM-43746 - Each SMI support device must be identified using its correct physical address, as defined in the electrical schematics, and verified against its part number, revision level, and BOM entry

4 Host PC Rules

4.1 General Rules

SM-44054 - Host PC General Rules

1. The host PC shall support both fully automated and prototype production modes.

- In **fully automated mode**, the system shall provide API and CLI interfaces to allow sequence files to be executed remotely via SSH or triggered by the automated production line from the same host.
- In **prototype mode**, a graphical user interface (GUI) shall be developed to enable users to manually run sequence files.

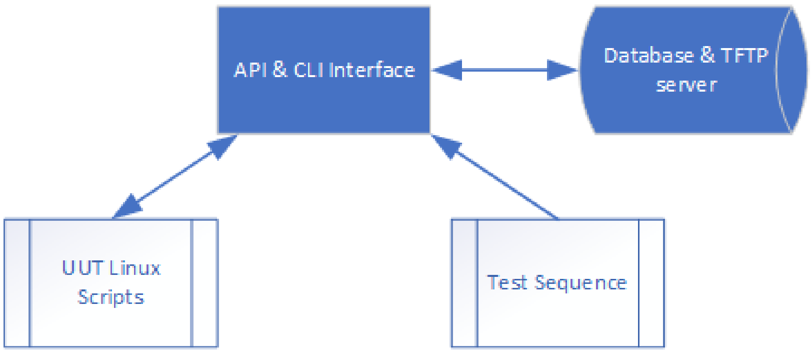
2. An automatic log shall be generated for each test run, containing the following information:

- Serial Number
- Start Time and Date
- End Time and Date
- Status – PASS or FAIL
- Fail Step – If failed, the specific test step name shall be recorded.
- Fail details – If failed, a brief description of the failure shall be included.
- Unique Field 1 t0 15 – These fields shall capture all relevant unique values (e.g., IS Key, FAZIT ID, etc.).
- Operator Name

3. For each test run, a full log capturing all data exchanged between the UUT and the host shall be automatically generated. The log shall be archived in a ZIP file, stored locally, and named using the station name, serial number, test result, and timestamp.
4. The host software shall manage all connected external tools, such as the power supply, test board, and other related equipment.
5. In both production modes, sequence files shall be automatically selected from a database based on the UUT part number, either by scanning the part number in prototype mode or by receiving it as a variable in automated mode.
6. Scripts and files requiring updates shall be automatically downloaded from a TFTP server to ensure they are up to date. All files shall be executed from the local host, not directly from the server.
7. The host PC shall run a Linux-based operating system.

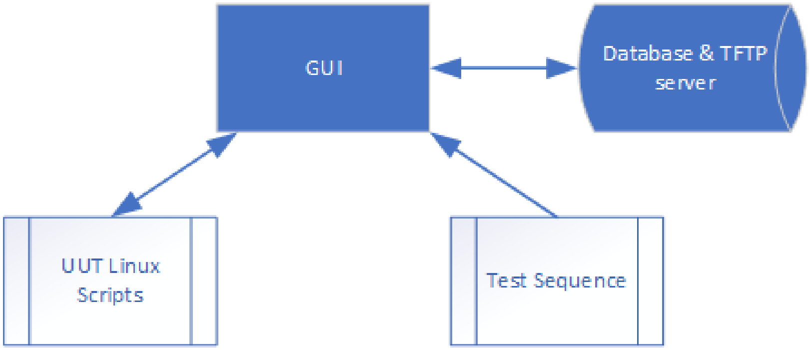
4.1.1 Automated Mode

SM-43741 - Automated mode process flow



4.1.2 Prototype Mode

SM-43755 - Prototype mode process flow



4.1.3 API & CLI Interface and GUI

SM-44057 - API & CLI Interface and GUI

1. The API, CLI, and GUI interfaces shall share the same underlying codebase and libraries to ensure consistent management of sequences in both automated and prototype modes. The API and CLI implementations shall invoke the same core functions used by the GUI.

2. Both operation modes shall support full sequence execution with the ability to halt the sequence immediately upon any failure (Stop of Fail).

4.1.4 Database

SM-44049 - Database

1. The host PC shall support both fully automated and prototype production modes

2. The database shall be based on SQL.
3. All files including sequence files, image files, scripts, and executables, shall be organized on the server using a file-based structure that separates general scripts from part-number-specific custom scripts.
4. Each host PC logs shall be copied to main server and updated at least each 4 hours.
5. The database structure shall be agreed with ME test engineering.

4.1.5 Test Flashing Sequence

SM-44052 - Test Flashing Sequence

1. The test sequence shall be defined as an ordered list of command steps. Each step shall include the following attributes:

- **Command**: The instruction or operation to be executed.
- **Expected Value(s)**: One or more expected output(s) against which the actual result will be evaluated.
- **Timeout**: Maximum allowed execution time (in milliseconds or seconds) before the step is considered failed.
- **Tolerance Range**: Minimum and maximum acceptable deviation from the expected value, where applicable.
- **Pass/Fail Logic**: Conditional branching based on the result of the step (e.g., Goto Step X on Pass, Goto Step Y on Fail).
- **Loop Control**: Optional loop parameters (e.g., iteration count or condition-based looping) for repeating the step or group of steps.

2. The test sequence shall be modular, composed of blocks that may include calls to subsequences or nested sub-blocks for hierarchical test organization.
3. The test sequence execution shall follow the steps below:

- **Prototype Mode (GUI):**Prompt the user to scan or input the Part Number, Serial Number, Operator Name, and other relevant identifiers.
- **Automated Mode (CLI/API):**Receive the Part Number, Serial Number, and other required variables as input parameters and perform validation.
- **External Tool Check:**Verify the connection status of all external tools, including reading and validating device ID codes.
- Power Supply Configuration:
  - Configure the power supply according to the test parameters.
  - Turn on the power based this document and power up rules.
  - Measure and validate the current against expected values as defined in the test sequence.
- **Flashing Procedure:** Execute the flashing process in accordance with the Flashing Order Rules o utlined in this document.
- **Post-Flash Verification:** Verify each flashed device to ensure correctness, based on the criteria defined in this document.
- **Shutdown:** Turn off the power supply after all test steps are completed.

5 General Requirements & Information

5.1 General Requirements

SM-44064 - General Requirements

1. The Contract Manufacturer (CM) shall prepare the Acceptance Test Procedure (ATP) for the Flashing program.

2. CM shall prepare a regression strategy plan and share it with ME Test Engineering team.
3. CM shall prepare a trouble shooting document and share it with ME Test Engineering team.
4. Flashing plan and report shall be provided using the specified document format.